4.1. Soups with two threat models

CIFAR-10. We explore the effect of interpolating two models robust to different ℓ_p-norm bounded attacks for $p \in \{\infty, 2, 1\}$. We consider classifiers with WIDERESNET-28-10 [49] architecture trained on CIFAR-10. For every threat model, we first train a robust classifier with adversarial training from random initialization. Then, we fine-tune the resulting model with adversarial training on each of the other threat models for 10 epochs. In Fig. 1 we show, for each pair of threat models (Δ_p, Δ_q), the trade-off of robust accuracy w.r.t. ℓ_p and ℓ_q for the soups

$$w \cdot \boldsymbol{\theta}_p + (1-w) \cdot \boldsymbol{\theta}_{p \to q} \quad \text{for} \quad w \in [0, 1]$$

and symmetrically

$$w \cdot \boldsymbol{\theta}_q + (1-w) \cdot \boldsymbol{\theta}_{q \to p} \quad \text{for} \quad w \in [0, 1].$$

Interpolating the parameters of models trained with a single ℓ_p-norm controls the balance between the two types of robustness: for example, Fig. 1 (middle plot) shows that moving from $\boldsymbol{\theta}_\infty$ to $\boldsymbol{\theta}_{\infty \to 1}$ (blue curve), i.e. decreasing w from 1 to 0 in the corresponding soup, progressively reduces the robust accuracy w.r.t. ℓ_∞ to improve robustness w.r.t. ℓ_1. Moreover, for similar threat models (i.e. the pairs (ℓ_2, ℓ_1) and (ℓ_2, ℓ_∞)) some intermediate networks are more robust than the extremes trained specifically for each threat model.

IMAGENET. We now fine-tune a VIT-B16 [8] robust w.r.t. ℓ_∞ on IMAGENET to the other threat models, including nominal training, for either 1/3, 1 or 3 epochs. Fig. 2 shows that interpolation of parameters is effective even in this setup, and allows to easily balance nominal and robust accuracy (fourth plot). Moreover, it is possible to create soups with two fine-tuned models, i.e. $\boldsymbol{\theta}_{\infty \to 2}$ and $\boldsymbol{\theta}_{\infty \to 1}$. Finally, increasing the number of fine-tuning steps yields better performance in the target threat model, which in turn generally leads to better soups.

Comparison to multi-norm robustness methods. Fig. 1 and Fig. 2 compare the performance of the model soups to that of models trained with methods for robustness in the union of multiple threat models. In particular, we show the results of MAX [42], which computes perturbations for each threat model and trains on that attaining the highest loss, and SAT [30], which samples uniformly at random for each training batch the attack to use. We train models with both methods for all pairs of threat models: as expected, MAX tends to focus on the most challenging threat model, sacrificing some robustness in the other one compared to SAT, since it uses only the strongest attack for each training point. When training for ℓ_∞ and ℓ_1, i.e. the extreme ℓ_p-norms we consider, MAX and SAT models lie above the front drawn by the model soups, hinting that the more diverse the attacks, the more difficult it is to preserve high robustness to both. In the other cases, both methods behave similarly to the soups. The main advantage given by interpolation is however the option of moving along the front without additional training cost: while one might tune the trade-off between the robustness in the two threat models in SAT, e.g. changing the sampling probability, this would still require training a new classifier for each setup.

4.2. Soups with three threat models

We here study the convex combination of three models with different types of robustness. For CIFAR-10 we create soups with each ℓ_p-robust classifier for $p \in \{\infty, 2, 1\}$ and its fine-tuned version into the other two threat models. We use the same models of the previous section, and sweep the interpolation weights $\boldsymbol{w} \in \mathbb{R}^3$ such that $w_i \in \{0, 0.2, 0.4, 0.6, 0.8, 1\}$ and $\sum_i w_i = 1$. In Fig. 3 and Fig. 10 (in the Appendix) we show clean accuracy (first column) and robust accuracy in ℓ_∞, ℓ_2 and ℓ_1 (second to fourth columns) and their union, when a point is considered robust only if it is such against all attacks (last column).

One can observe that, independently from the type of robustness of the base model (used as initialization for the fine-tuning), moving in the convex hull of the three parameters (e.g. $\boldsymbol{\theta}_\infty + \boldsymbol{\theta}_{\infty \to 2} + \boldsymbol{\theta}_{\infty \to 1}$ in Fig. 3) allows to smoothly control the trade-off of the different types of robustness. Interestingly, the highest ℓ_2-robustness is attained by intermediate soups, not by the model specifically fine-tuned w.r.t. ℓ_2, suggesting that model soups might even be beneficial to robustness in individual threat models (see more below). Moreover, the highest robustness in the union is given by interpolating only the models robust w.r.t. ℓ_∞ and ℓ_1, which is in line with the observation of [6] that training for the extreme norms is sufficient for robustness in the union of the three threat models. Although the robust accuracy in the union is lower than that of training simultaneously with all attacks e.g. with MAX (42.0% vs 47.2%), the model soups deliver competitive results without the need of co-training.

Finally, Fig. 4 shows that similar observations hold on IMAGENET, where we create soups fine-tuning classifiers robust w.r.t. ℓ_∞, with either RESNET-50 [16] or VIT-B16 as architecture, for 1 epoch.

4.3. Soups for improving individual threat model robustness

We notice in Fig. 1 and Fig. 2 that in a few cases the intermediate models obtain via parameters interpolation have higher robustness than the extreme ones, which are trained or fine-tuned with a specific threat model. As such, we analyze in more details the soups $w \cdot \boldsymbol{\theta}_\infty + (1-w) \cdot \boldsymbol{\theta}_{\infty \to 2}$, i.e. using the original classifier robust in ℓ_∞ and the one fine-tuned to ℓ_2, on both CIFAR-10 and IMAGENET. Fig. 5 shows the robust accuracy w.r.t. ℓ_∞ when varying the value of w: in both case the original model $\boldsymbol{\theta}_\infty$, highlighted in red, does not attain the best robustness. Interestingly, on

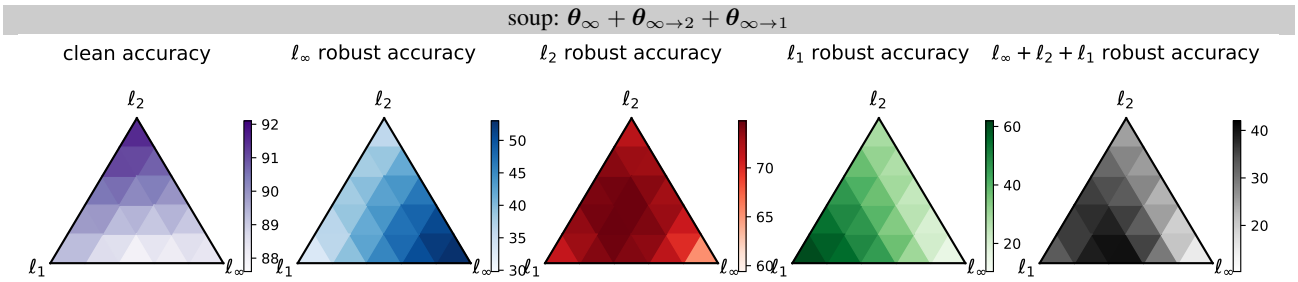

Figure 3. **Soups of three models on CIFAR-10:** we fine-tune the model robust w.r.t. ℓ_∞ (with WIDERESNET-28-10 architecture) to the other threat models for 10 epochs, and show clean accuracy (first column) and robust accuracy w.r.t. every threat model (second to fourth columns) and their union (last column) of the soups obtained as convex combinations of the three bases.

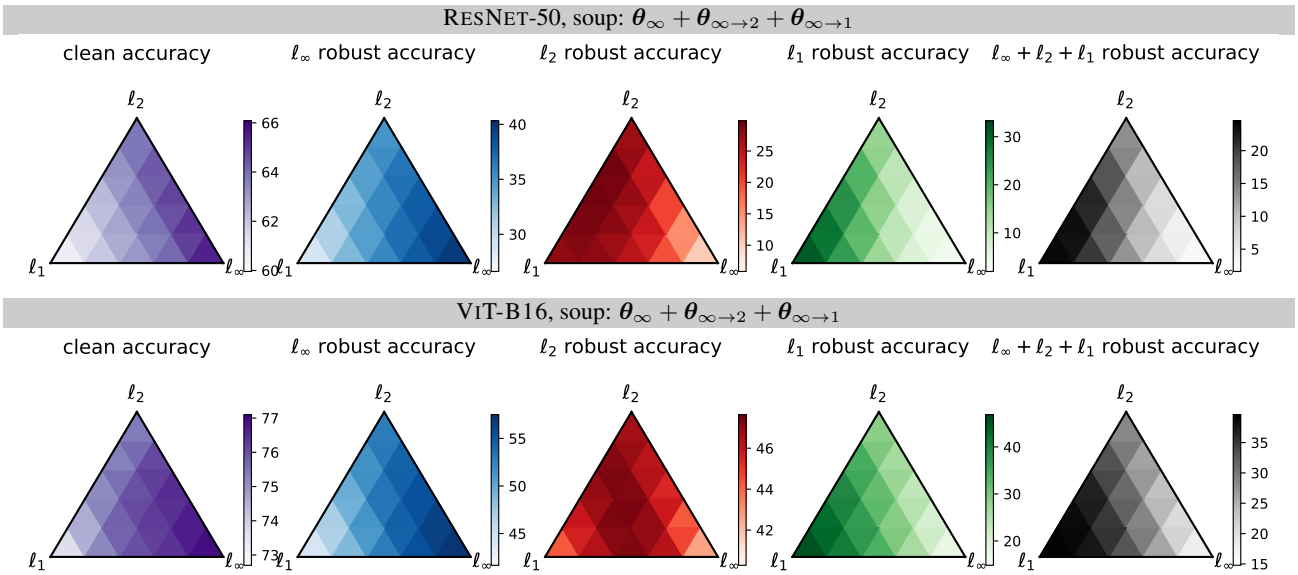

Figure 4. **Soups of three models on IMAGENET:** we fine-tune the classifiers, with RESNET-50 (top row) and VIT-B16 (bottom) as architecture, robust w.r.t. ℓ_∞ for 1 epoch to the other threat models, and show clean accuracy (first column) and robust accuracy in every threat model (second to fourth columns) and their union (last column) of the classifiers obtained as convex combinations of the three bases.

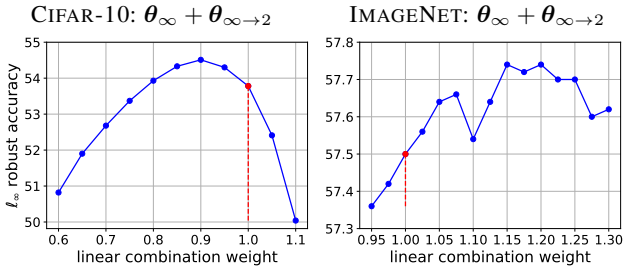

Figure 5. **Improvement on single threat models:** we show the robust accuracy w.r.t. ℓ_∞ for the soups $w \cdot \boldsymbol{\theta}_\infty + (1-w) \cdot \boldsymbol{\theta}_{\infty\to 2}$ for varied w. The original model $\boldsymbol{\theta}_\infty$ is highlighted in red.

CIFAR-10 the best soup is found with $w = 0.9$, while for IMAGENET with $w > 1$: this suggests that the model soups should not be constrained to the convex hull of the base models, and extrapolation can lead to improvement.

5. Soups for Distribution Shifts

Prior works [27] have shown that adversarial training w.r.t. an ℓ_p-norm is able to provide some improvement in the performance in presence of non-adversarial distribution shifts, e.g. the common corruptions of IMAGENET-C [18]. However, to see such gains it is necessary to carefully select the threat model, for example which ℓ_p-norm and size ϵ to bound the perturbations, to use during training. The experiments in Sec. 4 suggest that model soups of nominal and adversarially robust classifiers yield models with a variety of intermediate behaviors, and extrapolation might even deliver models which do not merely trade-off the robust-

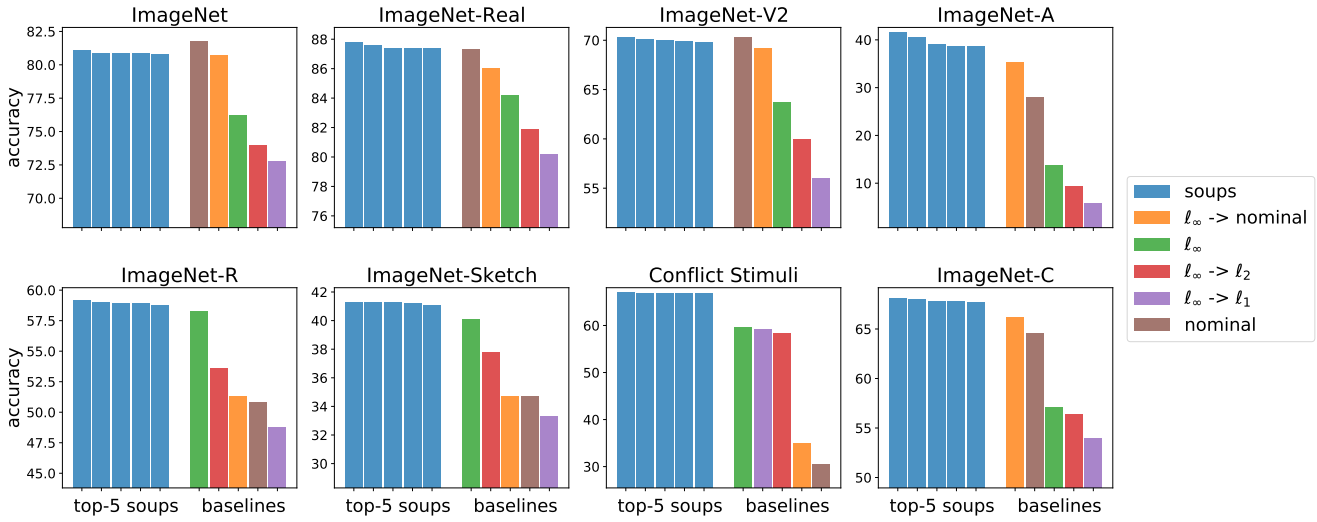

Figure 6. **Soups on IMAGENET variants:** for each dataset we plot the accuracy of the 5 best performing soups of the four base models $\boldsymbol{\theta}_\infty$, $\boldsymbol{\theta}_{\infty\to 2}$, $\boldsymbol{\theta}_{\infty\to 1}$ and $\boldsymbol{\theta}_{\infty\to\text{nominal}}$, and of the individual classifiers. Additionally, we show the performance of an independently trained nominal model. All models are evaluated on the 1000 points used for the grid search of the best soups.

ness of the initial classifiers but amplify it. This flexibility could suit adaptation to various distribution shifts: that is, the various corruption types might more closely resemble the geometry of different ℓ_p-balls or their union. Moreover, including a nominally fine-tuned model in the soup allows it to maintain, if necessary, high accuracy on the original dataset, which is often degraded by adversarial training [31] or test-time adaptation on shifted data [33].

5.1. Soups for IMAGENET variants

Setup. In the following, we use models soups consisting of robust VIT fine-tuned from ℓ_∞ to the other threat models, and one more VIT nominally fine-tuned for 100 epochs to obtain slightly higher accuracy on clean data. For shifts, we consider several variants of IMAGENET, providing a broad and diverse benchmark for our soups: IMAGENET-REAL [1], IMAGENET-V2 [36], IMAGENET-C [18], IMAGENET-A [20], IMAGENET-R [17], IMAGENET-SKETCH [44], and CONFLICT STIMULI [10]. We consider the setting of few-shot supervised adaptation, with a small set of labelled images from each shift, which we use to select the best soups.

Soup selection via grid search. Since evaluating the accuracy of many soups on the entirety of the datasets would be extremely expensive, we search for the best combination of the four models on a random subset of 1000 points from each dataset, with the exception of CONFLICT STIMULI for which all 1280 images are used (for IMAGENET-C we use all corruption types and severities, then aggregate the results). Restricting our search to a subset also serves our aim of finding a model soup which generalizes to the new distribution by only seeing a few examples. We evaluate all the possible affine combinations with weights in the range $[-0.4, 1.4]$ with granularity 0.2, which amounts to 460 models in total. In Fig. 6 we compare, for each dataset, the accuracy of the 5 best soups to that of each individual classifier used for creating the soups and of a nominal model trained independently: for all datasets apart from IMAGENET the top soups outperform the individual models. Moreover, we notice that the best individual model varies across datasets, indicating that it might be helpful to merge networks with different types of robustness.

Comparison to existing methods. Having selected the best soup for each variant (dataset-specific soups) on its chosen few-shot adaptation set, we evaluate the soup on the test set of the variant (results in Table 1). We also evaluate the model soup that attains the best average case accuracy over the adaptation sets for all variants (single soup), in order to gauge the best performance of a single, general model soup. We compare the soups to a nominal model, the ℓ_∞-robust classifier used in the soups, their ensemble, the Masked AutoEncoders of [15], AdvProp [47], Pyramid-AT [21], and the ensemble obtained by averaging the output (after softmax) of the four models included in the soups. Selecting the best soup on 1000 images of each datasets (results in the last row of Table 1) leads in 4 out of the 8 datasets to the best accuracy, and only slightly suboptimal values in the other cases: in particular, parameters interpolations is very effective on stronger shifts like IMAGENET-R and CONFLICT STIMULI, where it attains almost 8% better performance than the closest baseline. Unsurprisingly, it is more challenging to improve on datasets like IMAGENET-V2 which are very close to the original IMAGENET. Over-